

# HM1 — Control-Flow Diagrams (ISO 5807)

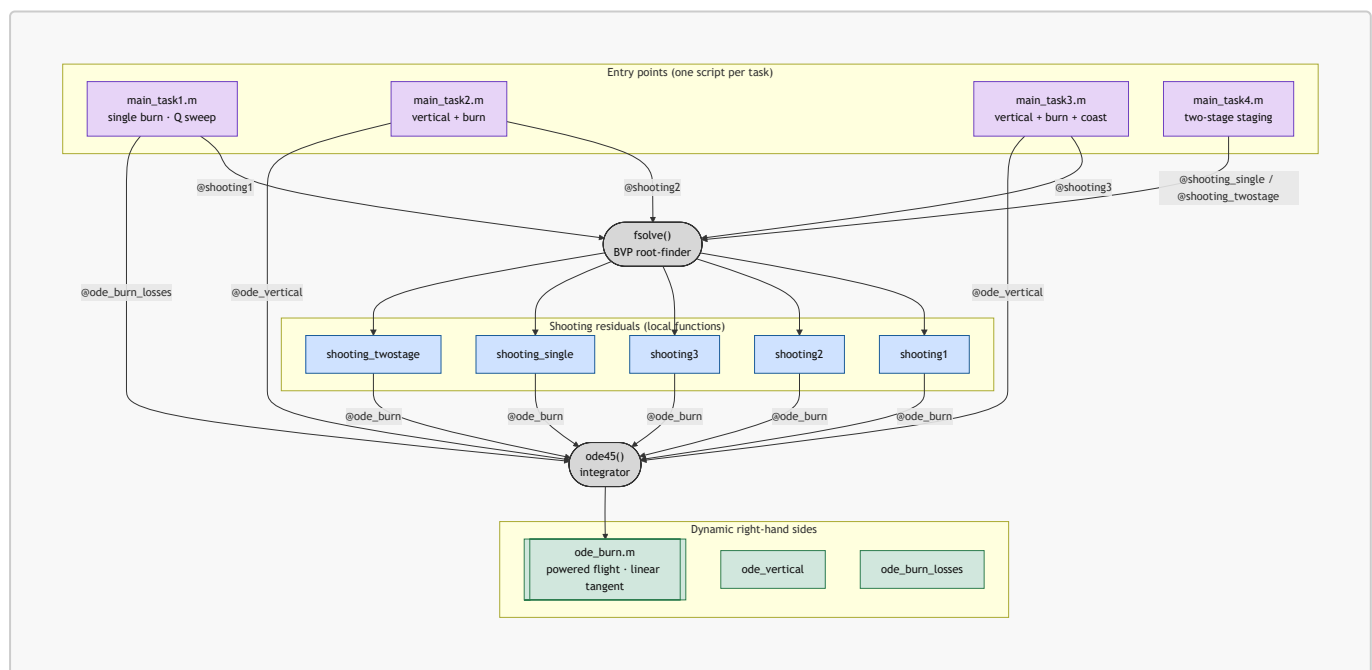
Detailed, color-coded step-by-step flowcharts of the four entry-point scripts and the shared routines in this folder, obtained by static reading of the source. Symbols follow **ISO 5807** (terminator, process, predefined process, decision, preparation, data I/O) and are mapped onto Mermaid node shapes — and colored by category — so the diagrams render natively on GitHub.

## Symbol & color legend

| ISO 5807 symbol          | Meaning                 | Mermaid node                     | Color  |
|--------------------------|-------------------------|----------------------------------|--------|
| Terminator (stadium)     | Start / End             | <code>([ ... ])</code>           | grey   |
| Preparation (hexagon)    | Setup / loop init       | <code>{{ ... }}</code>           | amber  |
| Process / predefined     | Operation / solver call | <code>[ ... ] · [[ ... ]]</code> | blue   |
| Decision (rhombus)       | Conditional branch      | <code>{ ... }</code>             | yellow |
| Data I/O (parallelogram) | print / figure / export | <code>[/ ... /]</code>           | green  |
| —                        | Error / abort path      | —                                | red    |

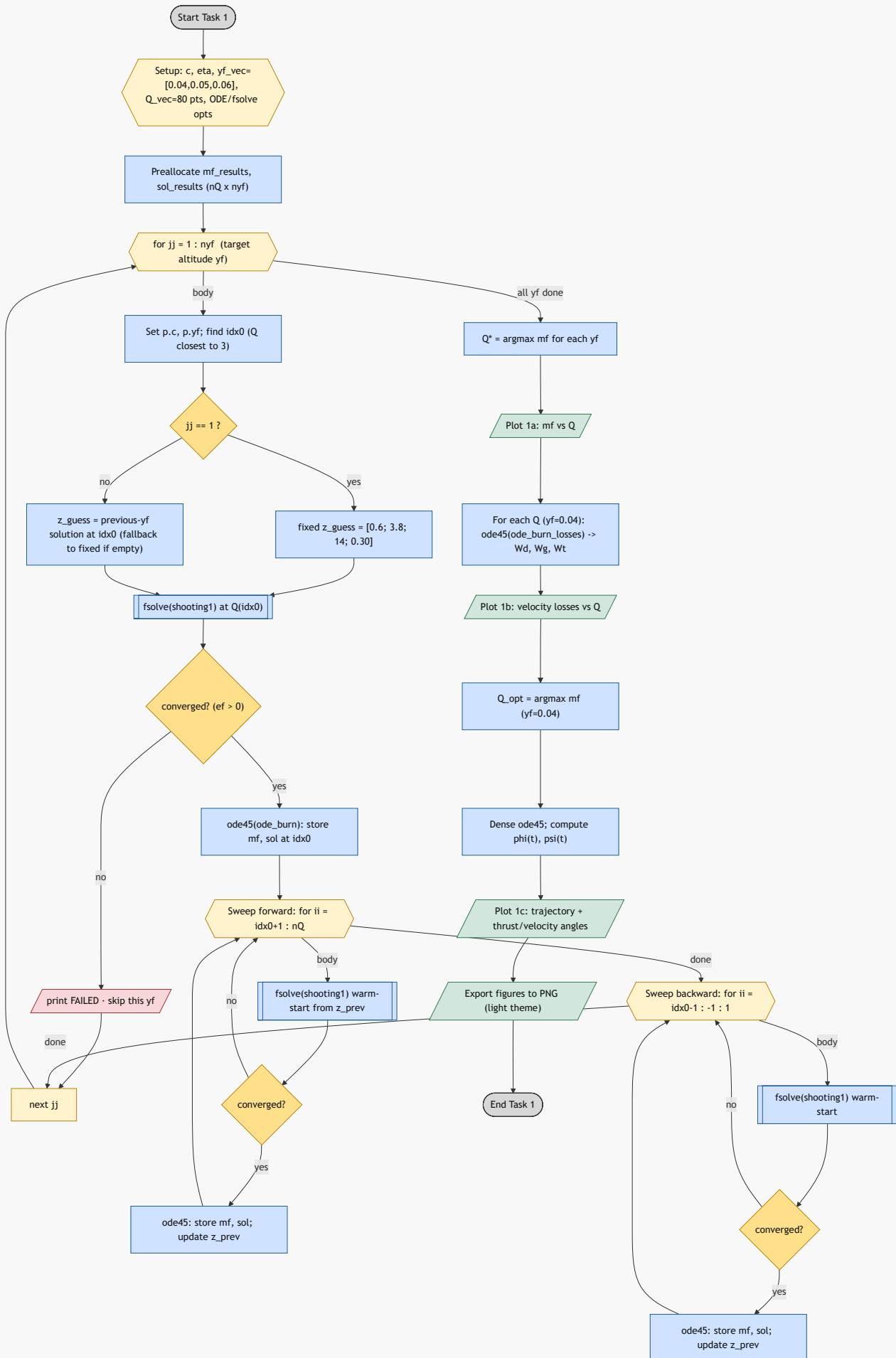
## 0 · Code architecture (call graph)

How the four entry points sit on the two numerical engines (**fsolve**, **ode45**) and the dynamic right-hand sides.



## 1 · Task 1 — Single burn arc, Q sweep with continuation

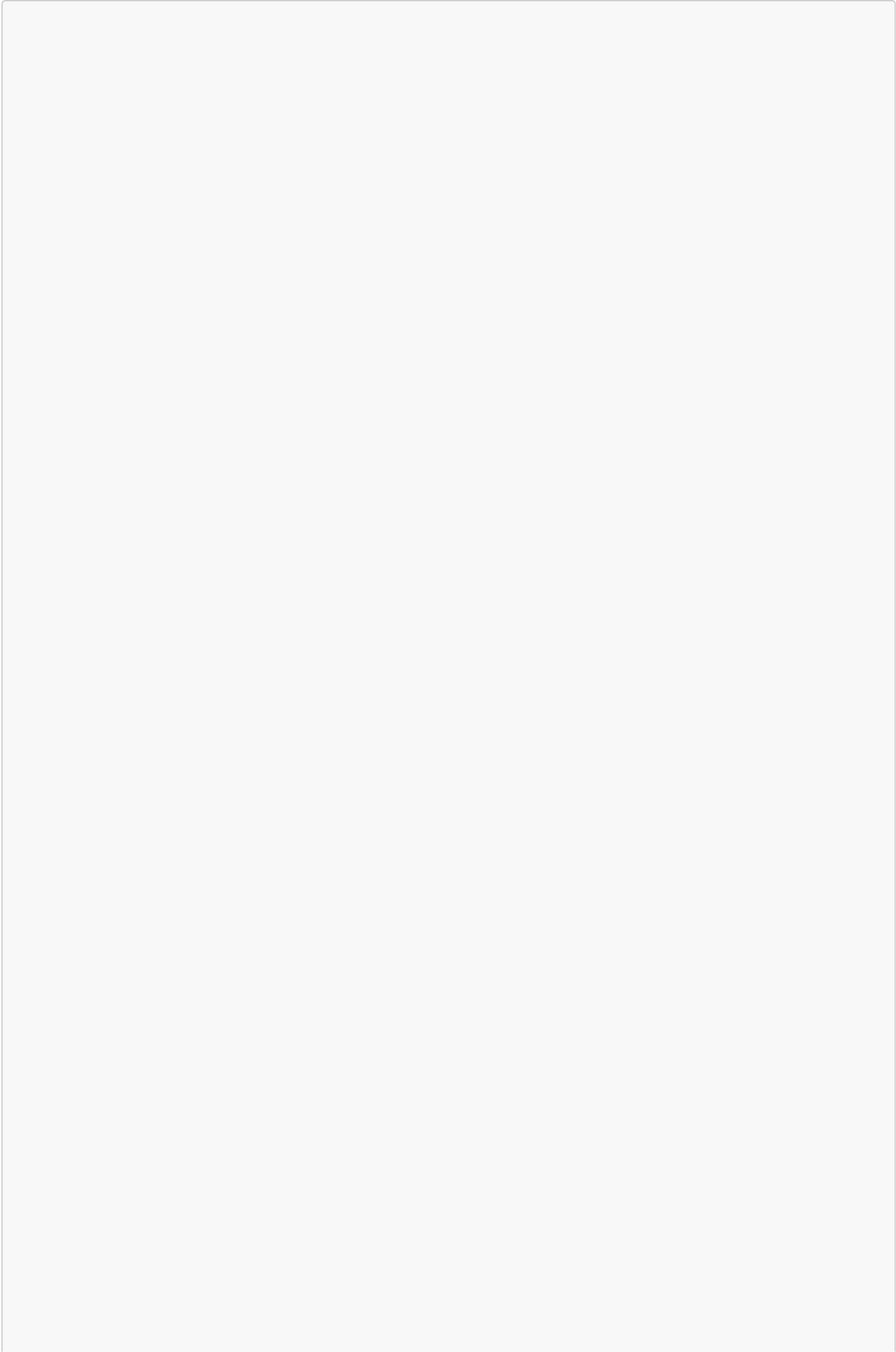
Bidirectional sweep (forward/backward) starting from  $Q \approx 3$  with warm-start of the previous solution; post-processing into three plots.

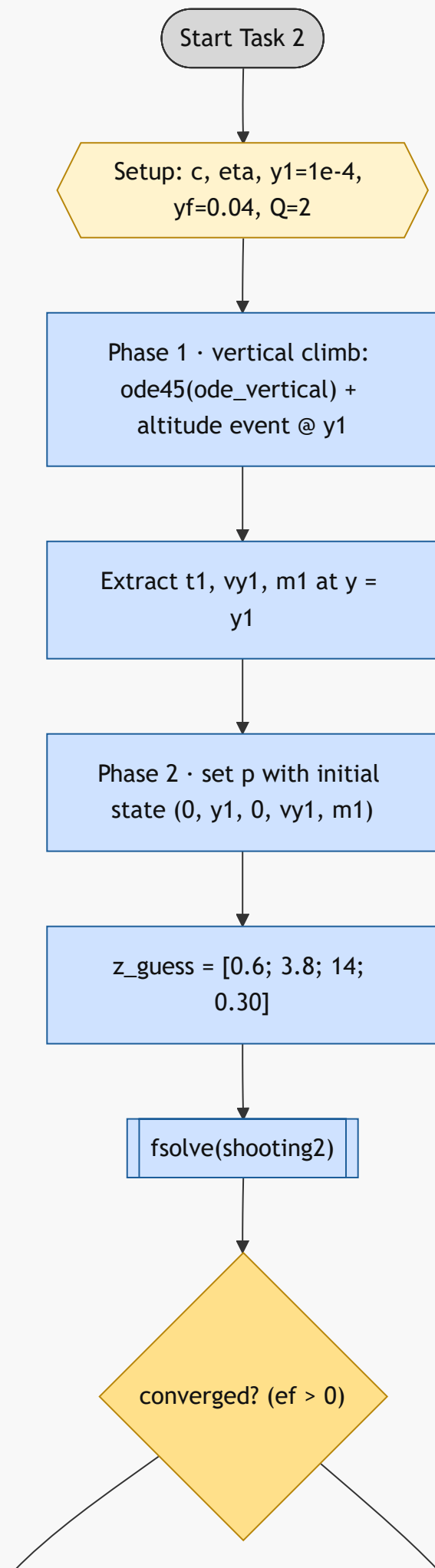


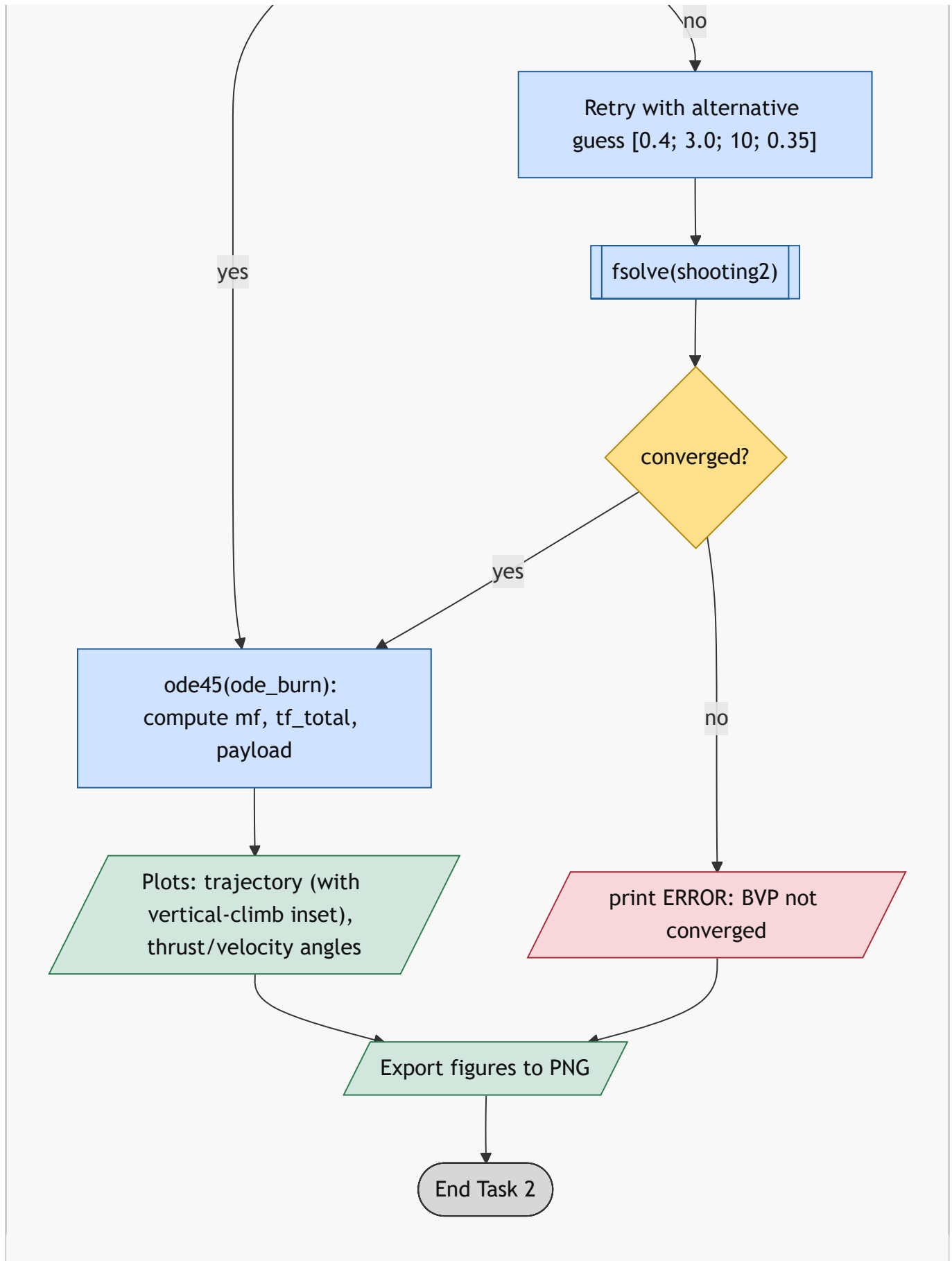
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## 2 · Task 2 — Vertical climb + optimal burn arc

Phase 1 with an altitude event; Phase 2 BVP with a fallback guess if the first solve fails.

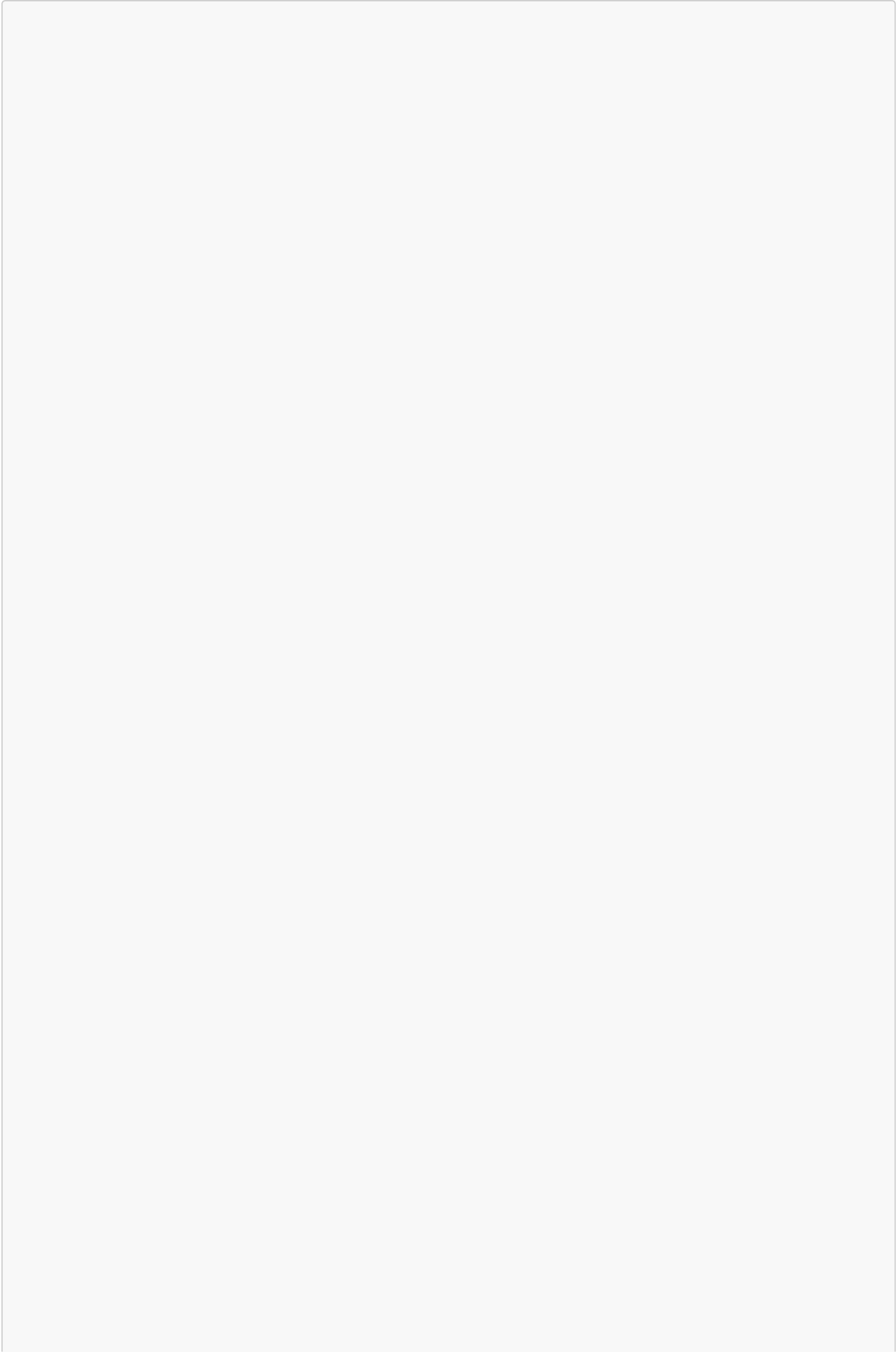




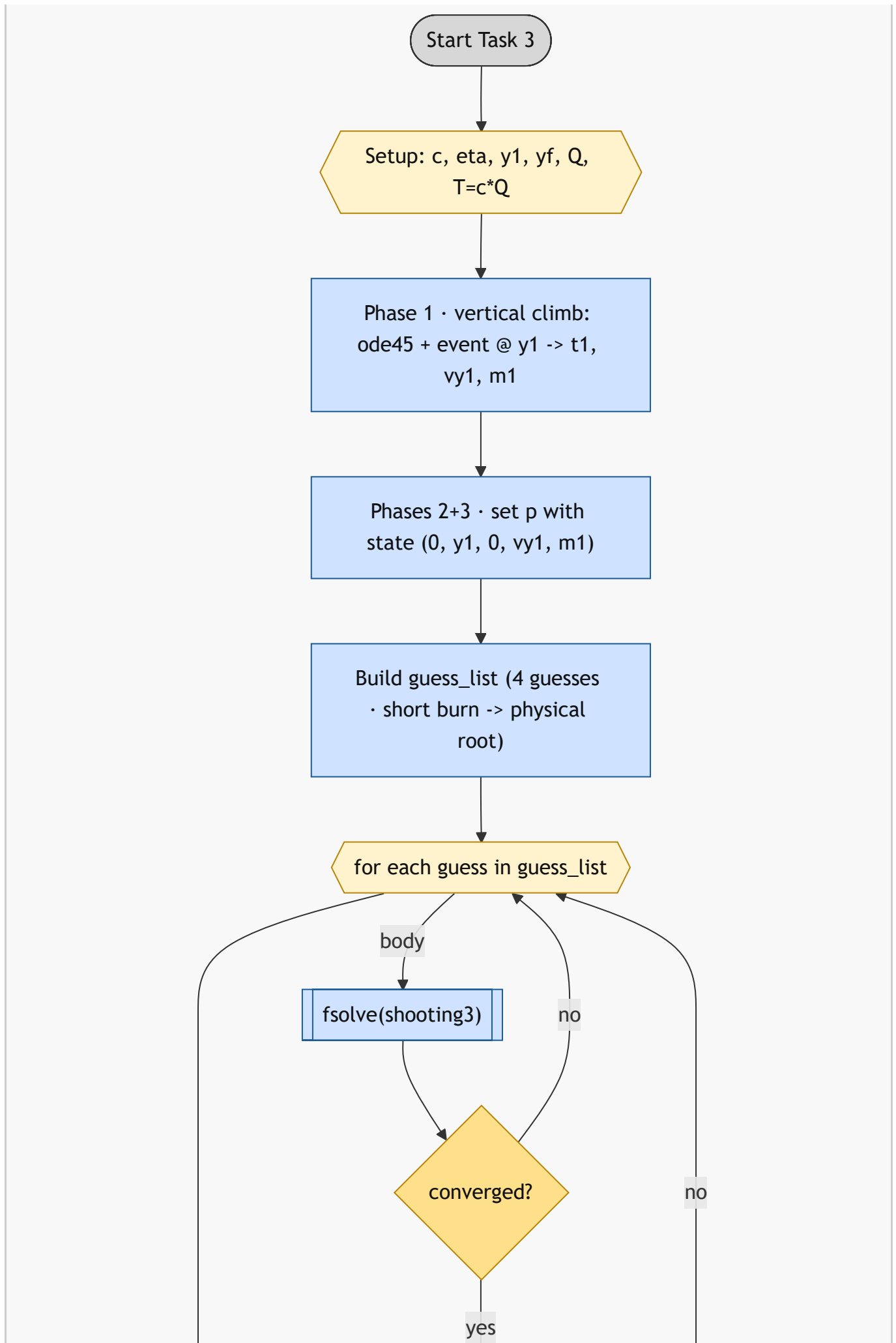


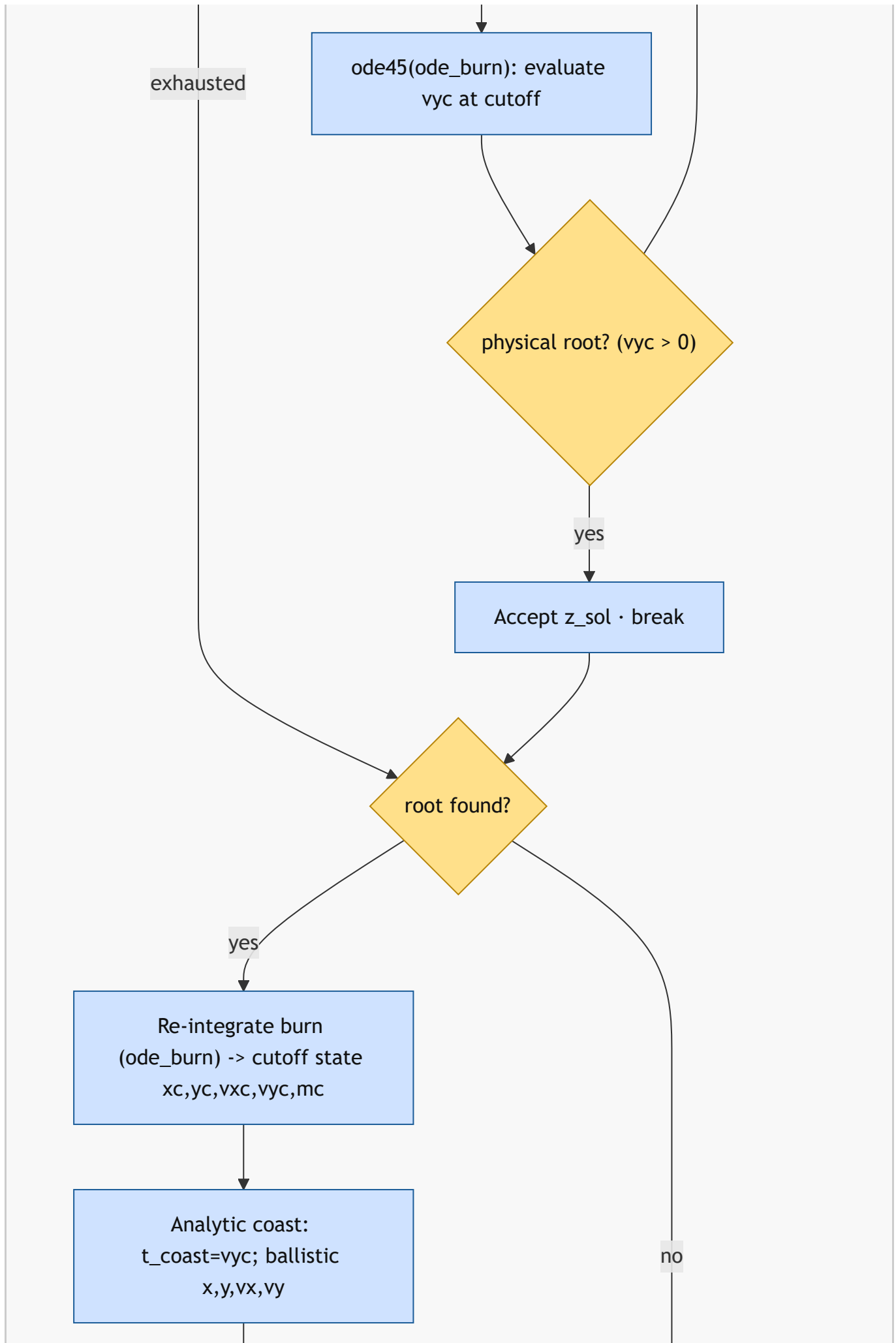
### 3 · Task 3 — Vertical + burn + coast (physical-root search)

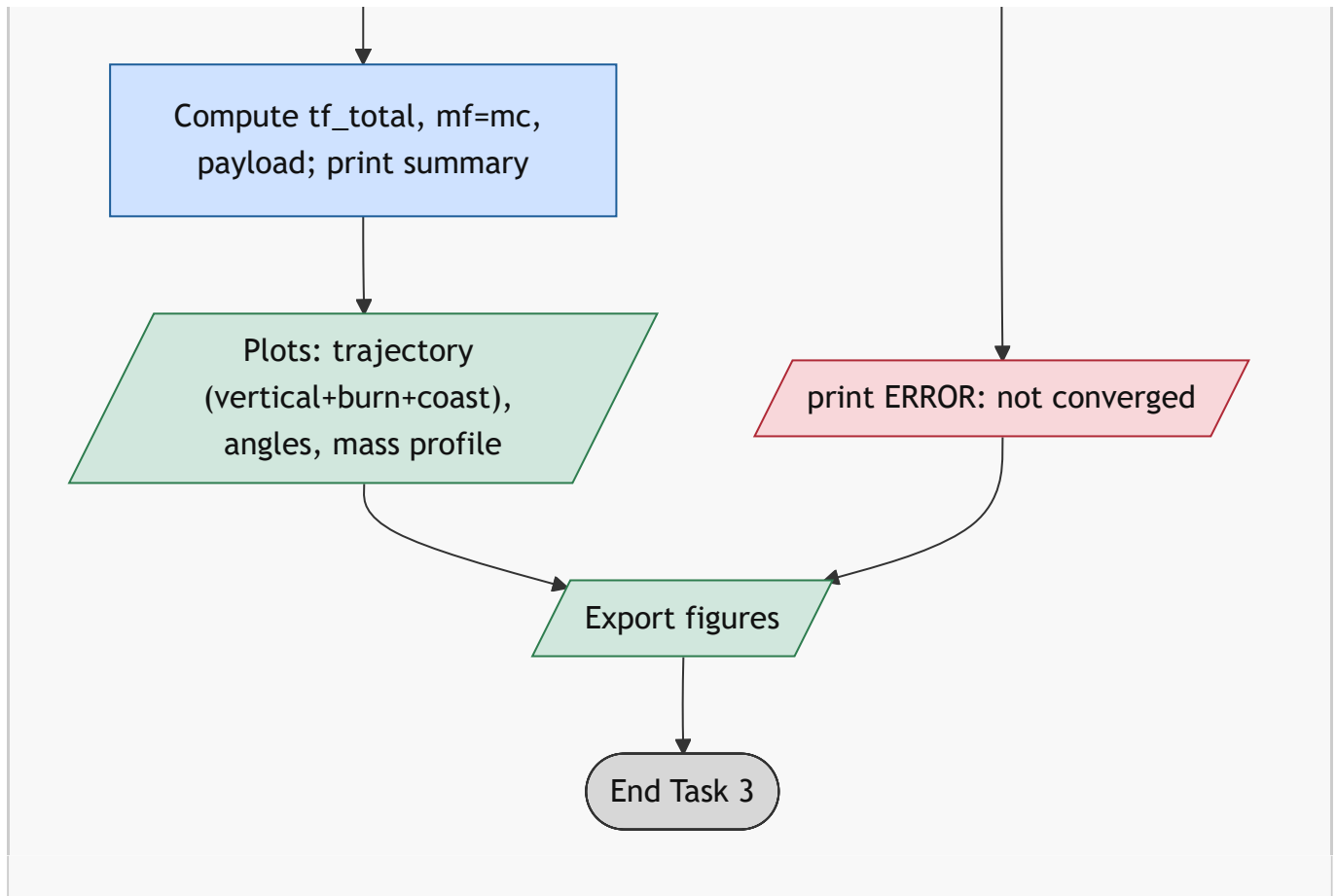
The four guesses are scanned, accepting the first root with  $v_{yc} > 0$  (physical ballistic coast); the coast phase is analytic.





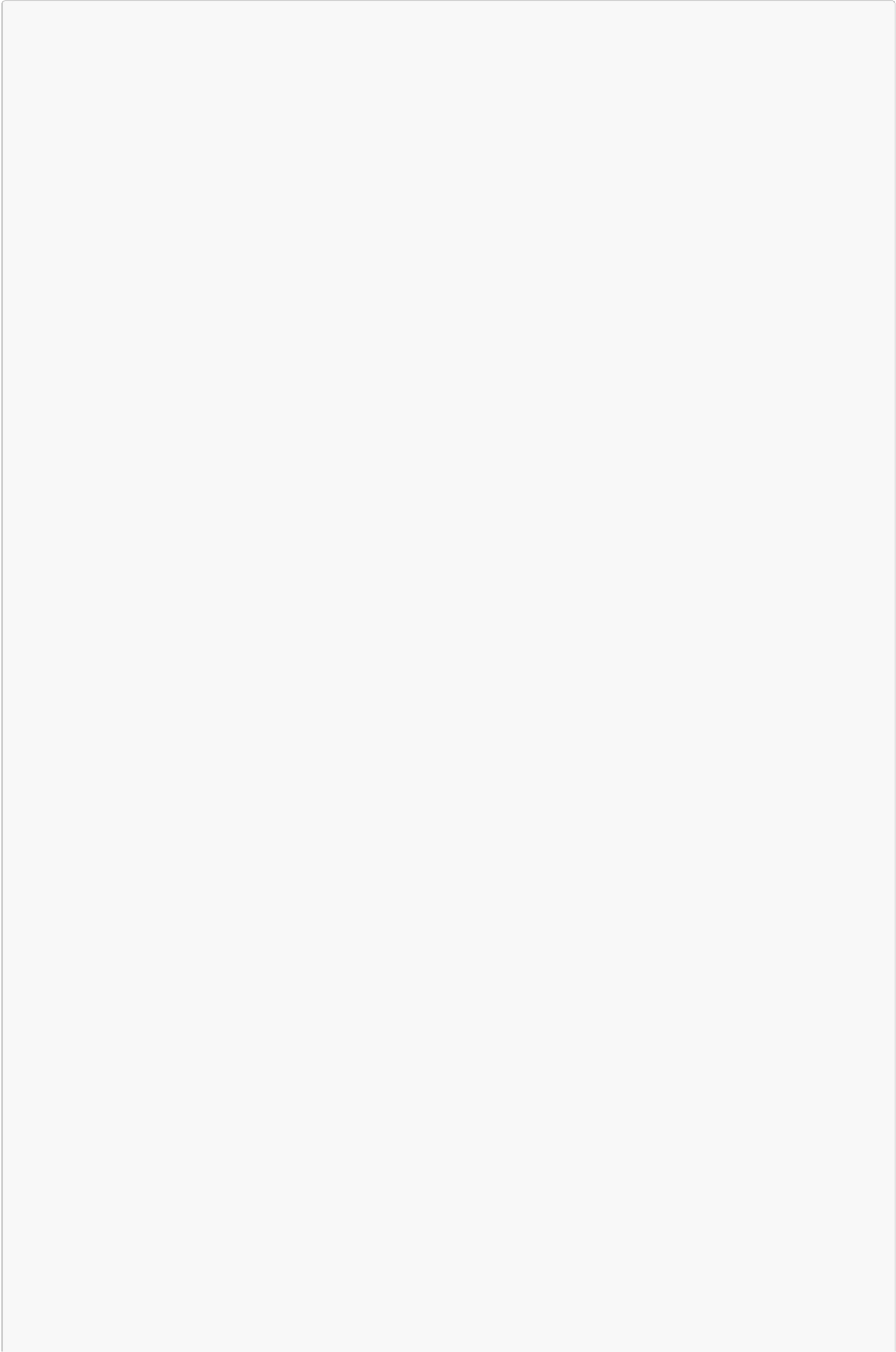


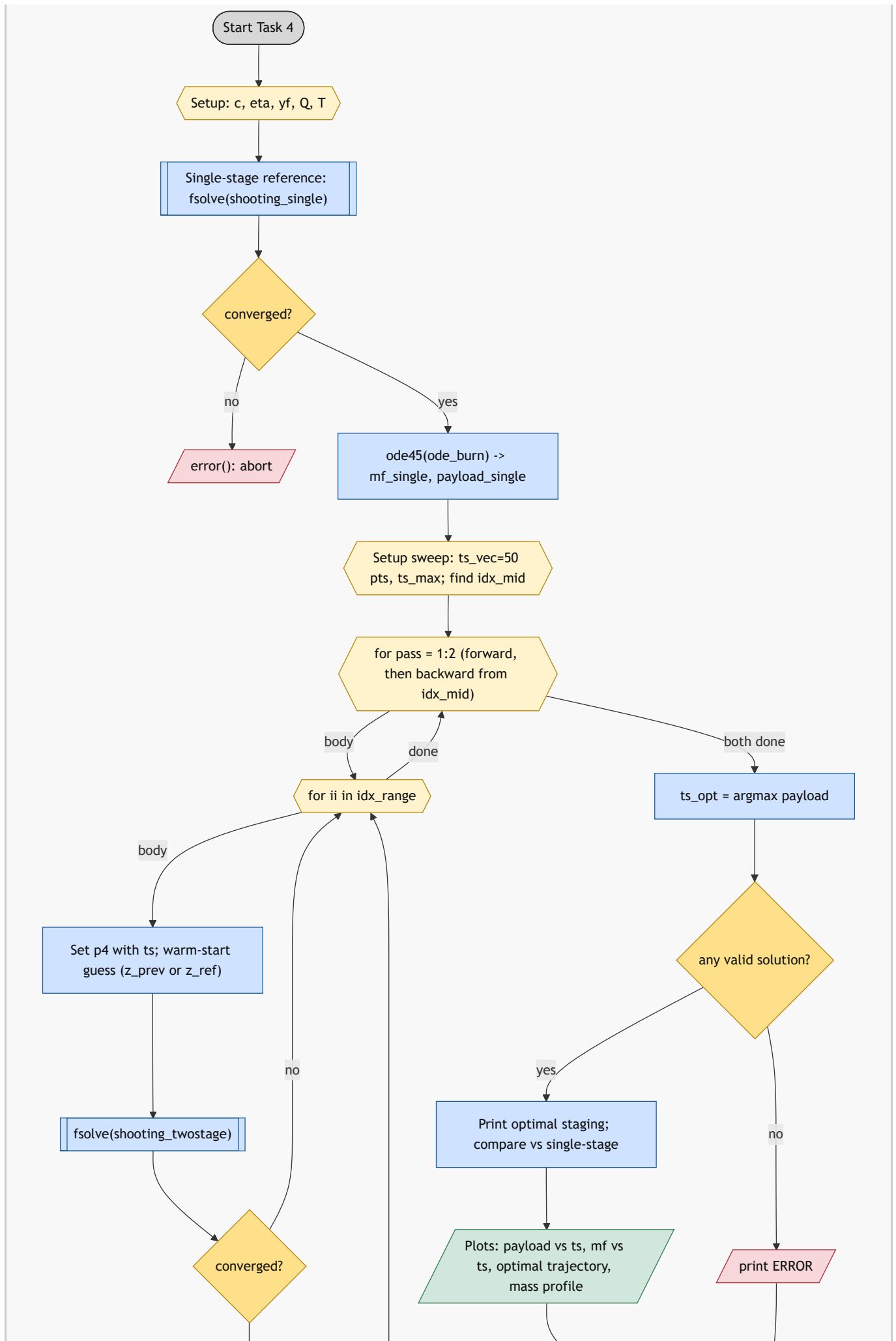


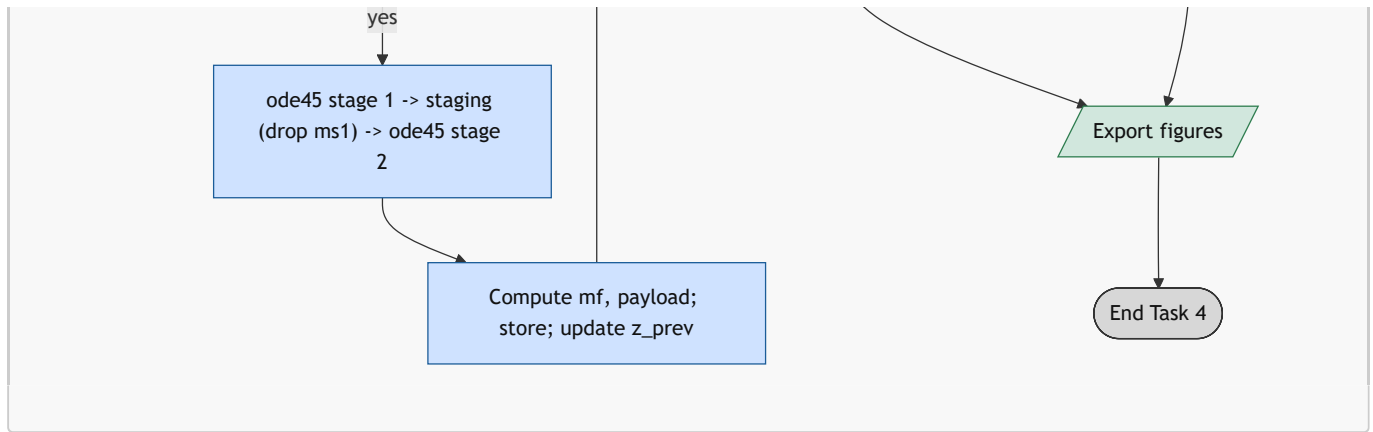


## 4 · Task 4 — Optimal staging (two stages)

Single-stage reference, then a bidirectional sweep of the staging time **ts** outward from the middle of the range.







## 5 · Shared subroutines

Left: the dynamic RHS (`ode_burn`, linear-tangent steering). Right: the common pattern of every shooting residual — a guard on `tf`, a `try/catch` around the integration, and the free-time condition  $H = 0$  imposed algebraically at the initial instant.



## Syntax error in text

mermaid version 11.16.0

**Note.** `shooting3` departs from the common pattern: 5 unknowns (`lam_m0` kept free) plus switching/coast conditions at cutoff instead of  $H = 0$  at `t0`. It is the only variant not reducible to the 4-unknown scheme above.